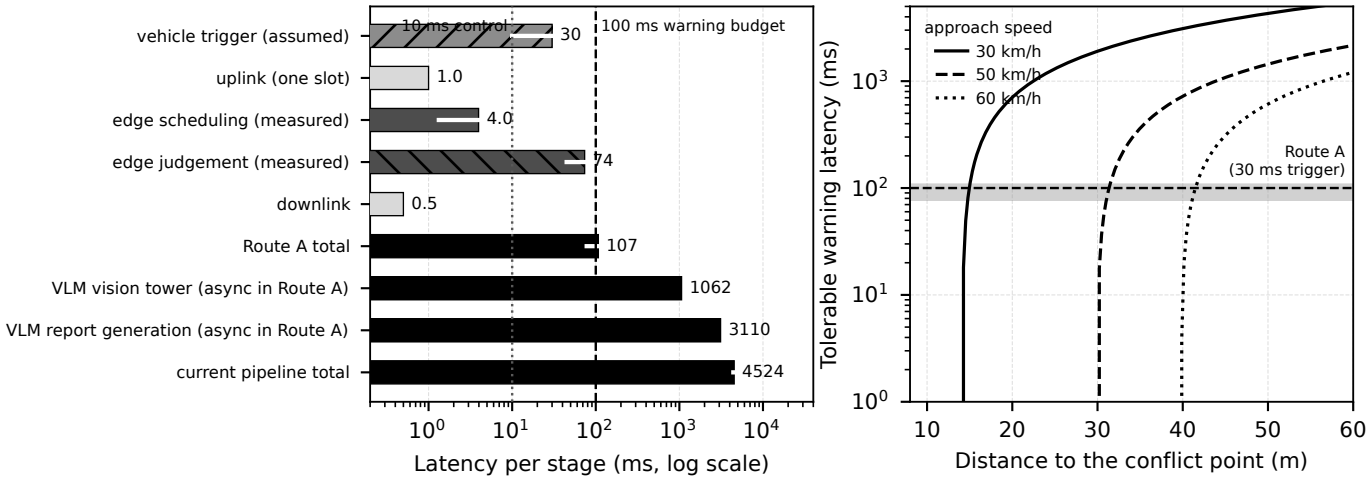
(a) Warning critical path
vehicle trigger (assumed)
uplink (one slot)
edge scheduling (measured)
edge judgement (measured)
downlink
Route A total
VLM vision tower (async in Route A)
VLM report generation (async in Route A)
current pipeline total
10 ms control
100 ms warning budget
30
1.0
4.0
74
0.5
107
1062
3110
4524
10^0
10^1
10^2
10^3
10^4
Latency per stage (ms, log scale)
(b) Timeliness margin
approach speed
30 km/h
50 km/h
60 km/h
Route A
(30 ms trigger)
Tolerable warning latency (ms)
10^0
10^1
10^2
10^3
10
20
30
40
50
60
Distance to the conflict point (m)